Textual Feature Construction under Reduced Context Affects Classifier Generalization in Short-Text News Classification

Group 8: Peiyan Chen (1317103) Hao Tian (1342982) Haozhou Du (1346881)

1. Introduction

Short-text inputs (≈5–50 words) provide limited contextual evidence, so a single missing content word can substantially shift the available signal. This project asks: *How does textual feature construction under reduced context affect the generalization of classifiers with different model complexity in short-text news topic classification?* We examine how performance changes as the input grows from a truncated headline to a headline with full short description, and whether simple, medium, and complex models respond differently to context reduction.

We use the News Category Dataset by Misra [1], restricted to the top 10 categories with 2,000 articles per class (20,000 instances). We construct four context levels: **L1** = first 5 headline words; **L2** = full headline; **L3** = headline + first 15 description words; **L4** = headline + full description. Three algorithms of increasing complexity are compared: Logistic Regression (LR), BiLSTM, and fine-tuned DeBERTa-base.

2. Literature Review

The News Category Dataset [1] pairs `headline` and `short_description` fields, allowing us to recombine them into controlled reduced-context inputs. Static word embeddings such as Word2Vec [5] assign each word a fixed vector and cannot adapt to local context, which is limiting for short text where topic signal depends on neighbouring words. BERT [3] introduced contextual pretrained representations where each token embedding depends on its surrounding tokens; when BERT is used as a frozen feature extractor, mean pooling typically outperforms the raw `[CLS]` vector [4]. DeBERTa [2] extends BERT through *disentangled attention* — each token has separate content and relative-position vectors, with a four-way attention decomposition (content↔content, content↔position) — plus an *enhanced mask decoder* injecting absolute position only at the final decoding step. These choices are directly relevant to reduced-context classification: short headlines contain few content words, so contextual and positional interactions carry disproportionate signal. We use `microsoft/deberta-base` as the post-2016 research-paper model.

3. Methods

3.1 Feature construction

We treat feature construction at two levels: the primary variable is the amount of textual context (L1–L4); the secondary variable is textual representation.

For sparse LR variants we concatenate word-level TF-IDF (1–2 grams, 10k max features) with character-level TF-IDF (3–5 character grams, 15k max features), yielding ∼25k sparse features. Word-level features capture topic-indicative terms; character-level features capture sub-word patterns useful for short or morphologically varied text. For the TF-IDF+POS variant we additionally add 10 hand-crafted POS features extracted with NLTK: counts of nouns, verbs, adjectives, adverbs, their per-token ratios, total token count, and mean word length, tested as residual signal beyond lexical n-grams. For LR+BERT we use `bert-base-uncased` as a *frozen* feature extractor, mean-pooling token embeddings to obtain a 768-d sentence representation. For BiLSTM+Word2Vec, we train a 100-d skip-gram Word2Vec model on training text only within each fold; the per-fold vocabulary maps `<PAD>` to a zero vector and `<UNK>` to unseen tokens. For BiLSTM+BERT, we precompute 768-d frozen `bert-base-uncased` token-level embeddings (sequential, not collapsed) as input to the BiLSTM, isolating the contribution of contextual representations from end-to-end fine-tuning.

To prevent leakage, all data-dependent preprocessing (TF-IDF vectorizers, POS standardization, Word2Vec vocabulary and weights) is fitted only on the inner-training split during hyperparameter selection, refitted on the full outer-training partition after selection, and applied once to the held-out outer-test fold. Frozen BERT weights are deterministic pretrained features and are not updated before outer-test evaluation.

3.2 Algorithms

Simple — Logistic Regression. A linear multi-class classifier with the `liblinear` solver, used identically across TF-IDF, TF-IDF+POS, and BERT representations. **Medium — BiLSTM.** A single-layer bidirectional LSTM over either Word2Vec vectors or frozen BERT token embeddings, followed by mean pooling, dropout (0.3), and a linear classifier into 10 classes; trained with Adam. **Complex — DeBERTa-base** [2]. Fine-tuned with AdamW (weight decay 0.01, linear warmup 10%), batch size 32, maximum sequence length 96, 3 epochs. We use FP32 throughout to keep results numerically stable and consistent across GPU runtimes.

3.3 Cross-validation and evaluation

We implement stratified k-fold splitting from scratch (`make_stratified_folds`). All three algorithms use the same outer 10-fold splits seeded with `random_state=42`; because the data-loading code is byte-identical across the three notebooks, the held-out partitions are exactly the same 10 sets of 2,000 articles — the comparison in Table 1 is therefore *paired* across algorithms, not merely matched in distribution. Within each outer fold we run inner 3-fold stratified CV (different inner-seed per outer fold), select the hyperparameter with highest mean inner macro F1, refit on the full outer-training partition, and evaluate once on the held-out outer-test fold. Accuracy, macro F1, weighted F1, and per-class F1 are computed from scratch from predictions; we report mean and standard deviation over the 10 outer folds.

3.4 Hyperparameter grids

Each algorithm has one hyperparameter tuned via inner 3-fold CV: LR regularization $C \in \{0.1, 1, 10\}$ shared across all three representations for a controlled within-classifier comparison; BiLSTM hidden dimension $\in \{128, 256, 384\}$ for the frozen-BERT variant and $\{64, 128, 256\}$ for the Word2Vec variant (smaller because Word2Vec inputs are 100-d versus 768-d); DeBERTa learning rate $\in \{2\text{e-}5, 5\text{e-}5, 8\text{e-}5\}$. Selection frequencies and middle-value-criterion outcomes are reported in Section 4.4.

4. Results and Discussion

4.1 Feature construction within the linear classifier

Adding 10 hand-crafted POS features to TF-IDF gives a small but consistent improvement of +0.001 to +0.005 macro F1 across L1–L4 — shallow syntactic composition contains some residual signal, but most discriminative information is already captured by the ∼25k lexical and character n-gram features. Replacing TF-IDF+POS with frozen mean-pooled BERT gives a larger improvement of +0.005 to +0.011, with the gain widening at longer context. Critically, however, **the gain from changing representation (0.008–0.016 within one context level) is dwarfed by the gain from increasing context** (+0.180 macro F1 from L1 to L4 for LR+BERT). Under a linear classifier, missing textual evidence is the dominant bottleneck; better representations partially mitigate but do not eliminate this loss.

Table 1: Test performance across the four context levels using three evaluation metrics: Accuracy, Macro F1, and Weighted F1. Values are mean ± std for the 10 outer folds. **Bold** and *Italics* indicate the best and worst values in each column of each metric-context.

	Accuracy				Macro F1				Weighted F1			
Configuration	**L1**	**L2**	**L3**	**L4**	**L1**	**L2**	**L3**	**L4**	**L1**	**L2**	**L3**	**L4**
LR + TF-IDF	.562±.007	.690±.012	.716±.008	.736±.010	.562±.008	.690±.012	.716±.008	.735±.010	.562±.008	.690±.012	.716±.008	.735±.010
LR + TF-IDF + POS	.563±.008	.690±.011	.721±.010	.739±.011	.564±.008	.691±.011	.721±.010	.739±.011	.564±.008	.691±.011	.721±.010	.739±.011
LR + frozen BERT	.571±.010	.703±.009	.733±.008	.752±.010	.570±.010	.702±.009	.732±.009	.750±.010	.570±.010	.702±.009	.732±.009	.750±.010
BiLSTM + Word2Vec	*.241±.023*	*.352±.031*	*.454±.032*	*.500±.037*	*.188±.036*	*.325±.041*	*.435±.035*	*.489±.039*	*.188±.036*	*.325±.041*	*.435±.035*	*.489±.039*
BiLSTM + frozen BERT	.561±.008	.693±.008	.725±.013	.738±.011	.558±.009	.688±.009	.721±.013	.733±.012	.558±.009	.688±.009	.721±.013	.733±.012
DeBERTa fine-tuned	**.640±.009**	**.767±.011**	**.812±.009**	**.836±.007**	**.640±.010**	**.767±.011**	**.812±.009**	**.835±.007**	**.640±.010**	**.767±.011**	**.812±.009**	**.835±.007**

4.2 Overall Results and Model Comparison

Table 1 shows three evaluation measures. Since the sampled set is balanced and the outer folds all have the same class support, we will treat Weighted F1 as a methodological cross-check, rather than an independent ranking indicator, since it is the same score as Macro F1 up to rounding. Accuracy gives a different view: for BiLSTM+Word2Vec, it is much higher than Macro F1, suggesting uneven performance across classes. We use Macro F1 as the primary balanced multi-class metric in the main analysis below.

DeBERTa is uniformly strongest, with its advantage over the next-best system (LR+BERT) growing from +0.070 macro F1 at L1 to +0.085 at L4. Both models use pretrained contextual representations, but LR+BERT keeps BERT *frozen* while DeBERTa is fine-tuned end-to-end; the remaining gap therefore reflects task-specific transformer adaptation. Two mechanisms specific to DeBERTa [2] plausibly contribute. First, *disentangled attention* separates content and relative-position into independent vectors, so in 5-word inputs (L1) the model can reason about "which word is third" without entangling lexical and positional evidence, while frozen mean-pooled BERT collapses both into a single sentence vector. Second, fine-tuning lets attention patterns adapt to the news-classification objective, whereas frozen BERT relies on pooling features learned from masked language modelling — an objective with no incentive to separate news topics. The widening L1→L4 gap is consistent with these mechanisms: more tokens means more position-conditioned interactions for disentangled attention to exploit.

The expected simple–medium–complex ordering does *not* hold automatically. BiLSTM+Word2Vec performs substantially worse than LR+TF-IDF (gap ≈ 0.37 macro F1 at L1) despite being a more complex model class. Replacing Word2Vec with frozen BERT token embeddings while keeping the BiLSTM architecture fixed recovers performance to within 0.01–0.02 of LR+BERT at every level. This is quite intuitive: when applied to such short text tasks, the quality of word vector representations is far more crucial than the complexity of the sequence model. The input itself is very short, usually only 5 - 50 word tokens, and it simply cannot fully utilize the ability of LSTM to capture long-range dependencies. The additional set of trainable parameters in BiLSTM actually increases the difficulty of training optimization, while the performance gain is completely insufficient.

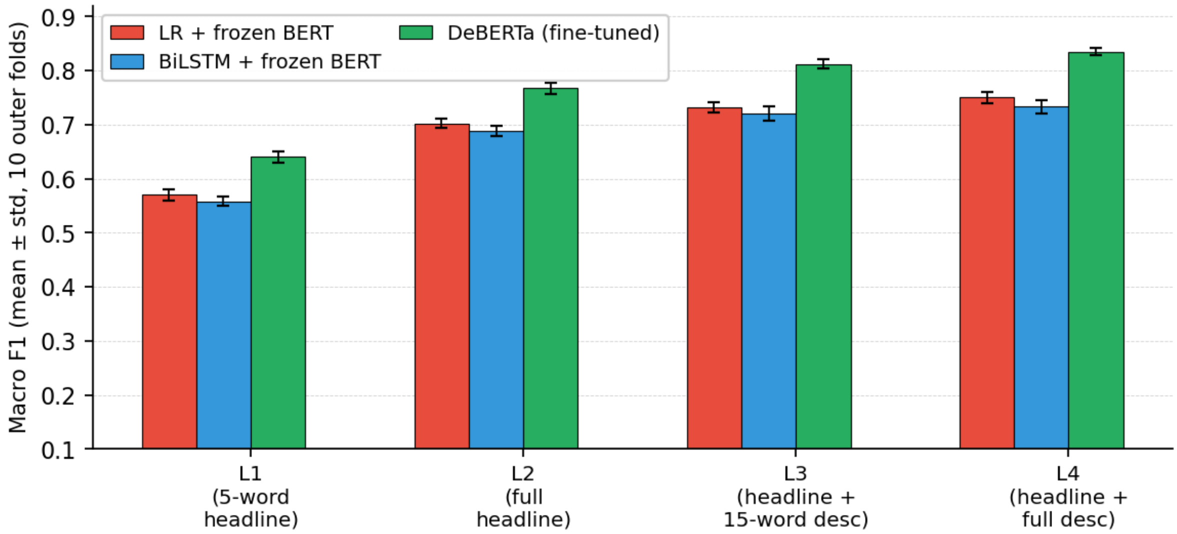

Figure 1: Macro F1 (mean ± std over 10 outer folds) across context levels for the three main configurations. DeBERTa is consistently strongest.

4.3 Impact of Context Reduction

The performance of the three core configurations monotonically improves from L1 to L4. The performance of LR+BERT increases by 0.180, that of BiLSTM+BERT by 0.175, and that of DeBERTa by 0.195. The BiLSTM+Word2Vec ablation achieves the largest gain, reaching 0.301. However, the overall F1 score of this model is relatively low. Due to its weak representation, it benefits more from added context. The macro F1 score of DeBERTa at the L1 level is only 0.640. Simply increasing the model complexity is insufficient to offset the performance loss caused by the significant lack of context.

A more refined pattern is that, across all configurations, gains decelerate as context grows, with the L3→L4 step the smallest. For DeBERTa the per-step gains are +0.127 (L1→L2), +0.045 (L2→L3), and +0.023 (L3→L4); the same shape holds for LR+BERT (+0.132, +0.030, +0.018) and BiLSTM+BERT (+0.130, +0.033, +0.012). The L3→L4 gain is consistently less than half of L2→L3, indicating **diminishing returns once a full headline plus the first 15 description words are available**. Most discriminative semantic signal is front-loaded into the lead of the short description — a property of journalistic writing rather than of any classifier — so the marginal value of further description text shrinks.

4.4 Hyperparameter selection behaviour

The middle-value criterion is satisfied for two of three algorithms. DeBERTa selected the middle learning rate 5e-5 in 33/40 outer-fold selections (with 2e-5 in 7/40 and 8e-5 never chosen), placing the optimum at the high end of the standard DeBERTa fine-tuning range. BiLSTM+BERT selected the middle hidden dimension 256 in 20/40, with 384 at 17/40 and 128 at 3/40, indicating moderate recurrent capacity is sufficient once strong contextual token features are available. LR+TF-IDF and LR+TF-IDF+POS each selected the middle $C = 1$ in all 40 selections. LR+frozen BERT, however, selected the boundary $C = 0.1$ in all 40 selections, indicating dense 768-d BERT embeddings require stronger regularization than sparse TF-IDF; we kept the shared grid for fair representation comparison rather than re-centering it, and flag this as a single-configuration deviation from the middle-value heuristic. Across all 120 LR selections, $C = 1$ remains modal (80/120, 66.7%).

4.5 Per-class structural difficulty

Figure 2 reveals a strong stratification across classes invisible in macro F1 alone. For DeBERTa, `HEALTHY LIVING` (0.92), `QUEER VOICES` (0.90), and `PARENTING` (0.90) saturate near the ceiling at L4, while `ENTERTAINMENT` (0.69) and `TRAVEL` (0.65) lag by 0.23–0.26, and they are the bottom-2 at every context level (L1: 0.475 and 0.436 respectively). For BiLSTM+Word2Vec — the architecture at the opposite extreme of our representation-quality spectrum — the same `TRAVEL`/`ENTERTAINMENT` pair forms the bottom-2 at L3 and L4 (at L1–L2, Word2Vec's macro F1 is too low for per-class rankings to be informative). The convergence of two architectures spanning fine-tuned contextual transformers and static word embeddings indicates the difficulty is **structural in the dataset, not architectural**. Because all classes are balanced to 2,000 samples, the cause is not class imbalance but **lexical breadth**: `ENTERTAINMENT` spans film, music, TV, celebrity, and award news with no shared vocabulary core, and `TRAVEL` headlines overlap `FOOD & DRINK` (food tourism), `STYLE & BEAUTY` (destination styling), and `PARENTING` (family travel). High-F1 classes carry tight lexical signatures — medical terminology, LGBTQ+ vocabulary, and family-life words respectively. This caps the L4 macro F1 ceiling near 0.84 regardless of model class.

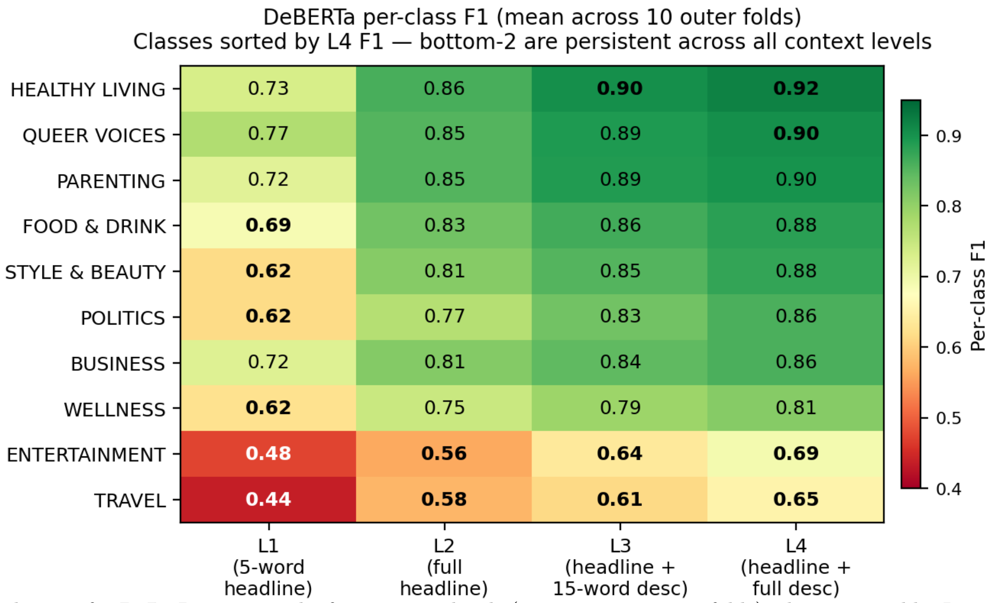

Figure 2: Per-class F1 for DeBERTa across the four context levels (mean over 10 outer folds), classes sorted by L4 F1. The bottom-2 classes (`ENTERTAINMENT`, `TRAVEL`) are persistent at every level for DeBERTa and form the bottom-2 for BiLSTM+Word2Vec at L3 and L4 as well — the structural ceiling is tied to label semantics rather than to architecture.

4.6 Limitations and alternative approaches

Limitations. For computational feasibility under the nested 10×3 CV protocol, BiLSTM+BERT was trained for one epoch and BiLSTM+Word2Vec for two. Thus, the BiLSTM comparison is conditional on light training; more epochs might narrow the gap to LR+BERT, although the Word2Vec–BERT representation gap is large across all context levels. DeBERTa was fine-tuned in FP32 to avoid mixed-precision artifacts at the cost of wall-clock time. Early stopping was not used because the inner validation split was already used for learning-rate selection.

Alternatives. We considered: (a) Linear SVM instead of LR, but it is near-equivalent to TF-IDF+LR here and lacks LR's calibrated probabilities for per-class analysis; (b) sentence-pooled frozen BERT as BiLSTM input, but this removes the token sequence the BiLSTM is intended to model; (c) classifier-head-only DeBERTa tuning to isolate the value of task-specific attention adaptation, but the compute cost under nested CV was prohibitive; and (d) full article body or temporal fields, which would move beyond the short-text scope.

5. Conclusion

We studied how textual feature construction under reduced context affects classifier generalization in short-text news classification, using from-scratch nested 10×3 CV with identical outer folds across three model-complexity classes. Three findings emerge. First, context helps monotonically but with diminishing returns: all methods improve from L1 to L4, while the L3→L4 gain is consistently smaller than L2→L3, suggesting that most signal is front-loaded in the short description. Second, DeBERTa is consistently strongest (macro F1 0.640–0.835), and its widening advantage over frozen-BERT LR suggests that fine-tuned contextual attention better exploits richer context. Third, representation quality dominates architecture for short text: BiLSTM+Word2Vec underperforms LR+TF-IDF, while BiLSTM+frozen BERT recovers to LR+BERT levels. Per-class analysis shows that the L4 ceiling near 0.84 is driven by broad labels such as Entertainment and Travel. Overall, under severe truncation, performance is limited by missing textual evidence rather than by classifier capacity.

Bibliography

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